

Brian Hill

Haymarket, VA | 703-946-1545 | hillbr20@gmail.com | brianchill.us/
[Github @TrojanPinata](#) | [Codeberg @Trojan_Pinata](#) | [LinkedIn: brian-hill-0082161a3](#)



SUMMARY

Software developer well-versed in embedded systems and microcontroller programming and hardware design. Driven to implement robust, real-world solutions applicable to a wide range of situations while focusing on meeting and exceeding customer expectations. Strong knowledge in both software development and hardware engineering, providing a blend of both skill sets while implementing complex solutions. Brings a unique outlook and energy to new situations and environments providing valued solutions to a variety of customers.

CLEARANCE

Active Public Trust Clearance

May 2023

WORK EXPERIENCE

Software Developer T1

Chantilly, VA

CACI International Inc. - Financial Management Business Transformation Project

May 2023 – Present

- Modified and developed SQL scripts for conversion between government financial systems.
- Developed Python-Selenium-based Robotic Process Automation (RPA) software for automated testing of government financial software (Momentum).
- Utilized government-based Agile approach to efficiently iterate on customer requirements.
- Tested and maintained current and legacy systems to validate performance and compliance with current systems.
- Analyzed client requirements and create necessary documentation for future development and integration.
- Developed Node-based GUI for Python backend in ReactJS.
- Analyzed legacy data to report issues which could effect conversion via automated workflows and tracking.
- Communicated directly with government customers for situational issues and problem solving.
- Served development owner of customer facing reports and the data processing pipeline that made them.

SKILLS & ATTRIBUTIONS

Languages: C/C++, Python, Rust, Java, SQL, JS/Web (ReactJS & Svelte), Bash, VHDL

Software: Git, Alteryx, PowerBI, Autodesk Inventor, KiCAD, PSpice, Docker/Kubernetes, Jenkins, SQL Developer, Nginx/Apache, Node, OpenCV, Linux, Virtual Machines

Development Focuses: Microcontrollers, Embedded Systems, Internet of Things, Autonomous Control Systems, Embedded Linux Systems, FPGAs

EDUCATION

George Mason University

Fairfax, VA

Computer Engineering, BS

- Notable Courses: Circuit Analysis I/II, Data Structures and Embedded Systems Programming, Continuous-Time Signals and Systems, Networking, Operating Systems, Computer Architecture, Embedded Systems, Linear Electronics, FPGA Design in VHDL, Microcontrollers, Internet of Things, Mobile Robots

Autonomous Blimp (LTAV) - Office of Naval Research Sponsored Capstone Project

<https://brianchill.us/projects/Blimp> - (More projects on brianchill.us)

- Developed unique shared Center of Gravity/Center of Buoyancy blimp from common components for high reproducibility while maintaining high maneuverability.
- Trained object detection model (YOLOv8) in high performance formats for reduced latency and to allow for accurate and fast detection.
- Designed and implemented functioning object detection, localization, and controls system in C++ and Python.
- Tested and deployed codebases on battery constrained hardware with four motor stabilization system and sensors.

AWARDS & CERTIFICATIONS

AWS Cloud Practitioner Accreditation | June 2023

GMU Tech Talent Awards | Multiple Years